AGENTIC AI

Capstone Projects

Three agentic AI projects: scope, objectives and agent assignments

CAPSTONE 1

Personalized Portfolio Advisor with What-If Simulation

CAPSTONE 2

Conversational Financial Statement Analyzer

CAPSTONE 3

Intelligent Invoice & Expense Reconciliation Agent

CAPSTONE 1

Personalized Portfolio Advisor with What-If Simulation

Brief

- Create a goal-based advisory agent where users enter financial goals, risk appetite, and current holdings, and receive a suggested allocation.
- The agent should stress-test the portfolio against scenarios (rate hike, market downturn, inflation), visualize projected outcomes in charts, and suggest rebalancing steps.
- Suitability disclaimers must be included. Bonus: multi-agent roles (risk profiler, allocator, simulator).

Objective

Deliver personalized, scenario-tested investment guidance with clear risk communication.

Learning takeaways

Goal-based financial planning logic, Monte Carlo / scenario simulation, data visualization, risk profiling, and embedding responsible-advice guardrails and disclaimers.

CAPSTONE 1 · PORTFOLIO ADVISOR

Agent roles and owners

Agent	Assigned To	Description
Data Agent	Pachha Harshvardhan	Ingests the portfolio CSV, fetches live prices via yfinance and news headlines per ticker; outputs a clean, structured dataset for everything downstream
Analysis Agent	Swapnil Bawankar	Computes portfolio metrics — returns, Sharpe ratio, sector allocation, concentration — and produces the analytics report the advisory chain reasons from
Risk Agent	Sujata Arak	Scores the investor's risk profile (conservative / moderate / aggressive) and evaluates whether current holdings match that tolerance
Advisory Agent	Mohit Kumar	Generates investment and rebalancing suggestions grounded in the RAG knowledge layer, tailored to the risk profile
QA Agent	Padmini N Dhawaskar	The compliance and suitability gate — checks every recommendation before it reaches the user, and can send it back for revision

CAPSTONE 2

Conversational Financial Statement Analyzer

Brief

- Create a GenAI tool where users upload company financials (P&L, balance sheet, cash flow) and ask questions conversationally.
- It should compute key ratios (liquidity, leverage, profitability), detect trends across periods, highlight red flags, and produce a plain-English narrative summary with citations back to the source figures.
- Bonus: compare two companies side by side.

Objective

Turn raw financial statements into grounded, queryable insight without manual spreadsheet work.

Learning takeaways

Retrieval-augmented generation (RAG) over documents, tool-use for reliable numeric computation, controlling hallucination on financial figures, grounding/citation techniques, and financial-ratio interpretation.

CAPSTONE 2 · FINANCIAL STATEMENT ANALYZER

Agent roles and owners

Agent	Assigned To	Description
Trend & Red Flag Agent	Vinayak Kulkarni	Computes period-over-period deltas across all ratios and evaluates the six red-flag patterns (accrual-quality gap, DSO climbing, margin compression, leverage stress, inventory build-up, liquidity deterioration)
Retrieval / Grounding Agent	Balaraju Kotturi	Indexes normalized statements and narrative notes in ChromaDB; on each question, assembles the grounded context block of computed facts plus relevant text
Narrative & Q&A Agent	Pramod Bhosale	The conversational surface — answers from grounded context only, with a mandatory citation on every figure, and declines when a number isn't in the statements
Comparison Agent	Sandip Shaw	Runs two companies side by side, surfacing fiscal-year-end or currency mismatches rather than comparing regardless

CAPSTONE 3

Intelligent Invoice & Expense Reconciliation Agent

Brief

- Build an agentic system that ingests invoices, receipts, and bank statements (PDF/image/CSV), performs a three-way match against purchase orders, and flags mismatches, duplicates, and anomalies.
- The agent should explain each exception in plain language and route uncertain cases to a human for approval.
- Bonus: auto-draft a follow-up email to the vendor for disputed items.

Objective

Automate transaction reconciliation and surface only the exceptions that need human attention.

Learning takeaways

Document parsing and OCR, structured data extraction from messy inputs, entity/transaction matching logic, anomaly detection, and designing human-in-the-loop approval flows.

CAPSTONE 3 · INVOICE & EXPENSE RECONCILIATION

Agent roles and owners

Agent	Assigned To	Description
Ingestion Agent	Vijay Vishwakarma	Accepts a mixed folder of PDFs, images and CSVs; routes by type (text-layer PDF, OCR for scans, pandas for CSV) and emits tagged raw text and tables
Extraction Agent	Bharatwaj Umashankar	Turns messy raw text into strict Pydantic records, handling missing fields, varied date formats and mangled OCR tables — attaches a per-field confidence score
Matching Agent	Radhika Khindri	Performs the three-way match (PO ↔ invoice ↔ receipt) with configurable tolerances and fuzzy vendor matching. Deterministic Python, no LLM — this has to be auditable
Anomaly Agent	Arun K	Independent of matching: detects duplicate invoices, statistical outliers against vendor history, and policy breaches like above-threshold amounts
Explanation Agent	Sanket Pusane	Turns each exception into a plain-language explanation a finance clerk can act on, citing the actual expected vs. found amounts
HITL Router	Sanat kumar	Decides auto-clear vs. human queue using confidence, severity and amount at risk; builds the approval queue and records decisions back into state
Correspondence Agent	Kapil Vermani	Drafts a professional vendor dispute email citing the PO number and specific discrepancy — always a draft for review, never auto-sent

CAPSTONE 4

AI Credit Risk & Loan Underwriting Assistant

Brief

- Build a decision-support agent that takes an applicant's profile and financial data and produces a structured risk assessment, a recommended decision, and a clear explanation of the reasoning.
- Users can simulate "what-if" changes (income, existing debt, loan amount), and fairness guardrails keep every decision auditable.
- Bonus: generate an applicant-facing explanation letter.

Objective

Produce explainable, auditable credit recommendations rather than a black-box score.

Learning takeaways

Structured reasoning and prompt design, explainable AI, bias/fairness checks, scenario simulation, and applying responsible-AI principles in a regulated domain.

CAPSTONE 4 · CREDIT RISK & LOAN UNDERWRITING

Agent roles and owners

Agent	Assigned To	Description
Intake Agent	Snigdha Meshram	Validates the applicant profile and financial data (income, debts, credit history, loan request) against a strict schema and flags missing or inconsistent fields before scoring
Metrics Agent	Shahaji Sargar	Computes underwriting ratios in deterministic Python: debt-to-income, payment-to-income, loan-to-value and credit utilization. No LLM, so every number is reproducible
Risk Assessment Agent	Abhishek Maurya	Applies credit policy rules with structured reasoning to assign a risk grade, listing each driving factor alongside the metric that supports it
Decision Agent	Asha Deshpande	Recommends approve, decline or refer, with conditions where relevant (lower amount, co-signer), and writes a structured decision record with a confidence level
What-If Simulator	Akshay Jain	Re-runs the pipeline when users change income, existing debt or loan amount, and shows how the risk grade and recommended decision shift
Fairness & Audit Agent	Shubham Tiwary	Keeps protected attributes and their proxies out of the reasoning, runs bias checks across applicant groups, and logs inputs, rules applied and outputs for audit
Explanation Letter Agent	?	Drafts an applicant-facing letter stating the decision and principal reasons in plain language — always a draft for review, never auto-sent